AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 13 · STUDENT LESSON

Types of Classification Error

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Read the reference. Read the prediction. Name the mistake.

Build a labelled count matrix without losing example IDs or confusing a positive class with a desirable outcome.

Keep one habit: Paired records → labelled cells → totals and error names → consequences and limits.

MAIN LESSON · SECTION 1

Which mistake did the classifier make?

> Keep each reference and prediction linked to its example ID.

Your learning goals

I can build a labelled **confusion matrix** from matched reference and **prediction** pairs.

I can distinguish **false positive**s from **false negative**s using a declared **positive class**.

I can check matrix totals and explain why equal accuracy can hide different errors.

Recall why a score alone loses evidence

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Why should a reproducible run save **prediction**s linked to example IDs rather than only an accuracy score?

Your end goal

By the end, I can construct a **confusion matrix** with declared axes and name both error types.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Rows Columns and Data Types. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 13: Types of Classification Error

False alarms and misses can have different costs.



THINKER MISSION

Sort 12 **prediction**s into four outcome boxes.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Read eight paired outcomes before counting

> Declare axes, class order and the positive class first.

What a confusion matrix records

A **confusion matrix** shows which labels a classifier mixes up. It counts examples after **prediction**s are made; it does not fit or repair the classifier.

Week 12 had reference/**prediction** pairs clear/clear, blocked/blocked, clear/blocked. A labelled table can preserve the type of that error as well as the total score.

The complete new eight-record evaluation

This week uses eight new fictional evaluation records from a frozen classifier. They are not extra **prediction**s from the Week 12 code. Each ID appears once with a reviewed reference and one **prediction**.

The grouped table below lists every ID. Keep reference and **prediction** paired by ID before counting.

Evaluation IDs	Reference	Prediction
V1 V2	clear	clear
V3	clear	blocked
V4 V5	blocked	clear

Evaluation IDs	Reference	Prediction
V6 V7 V8	blocked	blocked

The axis and positive-class convention

Use reference labels for rows and predicted labels for columns. On both axes the order is clear, then blocked. Declare blocked as the **positive class** and clear as negative.

Positive means the condition of interest, not good or safe. A missed blocked case and a false alarm are different errors.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use types of classification error to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

MAIN LESSON · SECTION 3

Trace one record into one matrix cell

> The code counts supplied outcomes; it does not train or repair a model.

Understand the tuple and lookup syntax

Read each tuple as (ID, reference, prediction). enumerate(labels) gives clear position0 and blocked position1; the dictionary comprehension builds those two lookups. counts starts as two rows of two zeros. The first lookup chooses the reference row and the second chooses the prediction column. +=1 adds one case to that cell. The loop unpacks each tuple into three names; it never guesses a reference label from a prediction.

Predict and run the complete counting script

```
pairs = [("V1", "clear", "clear"),
         ("V2", "clear", "clear"),
```

```
          ("V3", "clear", "blocked"),
          ("V4", "blocked", "clear"),
          ("V5", "blocked", "clear"),
          ("V6", "blocked", "blocked"),
          ("V7", "blocked", "blocked"),
          ("V8", "blocked", "blocked")]
labels = ["clear", "blocked"]
index = {label: i for i, label in enumerate(labels)}
counts = [[0, 0], [0, 0]]
for item_id, reference, prediction in pairs:
    counts[index[reference]][index[prediction]] += 1
print(counts)
correct = counts[0][0] + counts[1][1]
print(correct, len(pairs))
```

Explain the output and assumptions

index maps clear to 0 and blocked to 1. Each paired record adds one to its reference row and **prediction** column. The script prints [[2, 1], [2, 3]] then 5 8. It assumes unique IDs and only these two labels.

Check IDs and missing or unresolved references before counting; the code does not validate all those conditions.

Keep labelled cells, totals and the diagonal

Reference rows	Predicted clear	Predicted blocked
Reference clear	2	1
Reference blocked	2	3

The diagonal cells count correct **prediction**s: 2+3=5. Off-diagonal cells count mistakes: 1+2=3. Every record must contribute exactly once. This is a count matrix, not percentages or probabilities.

Keep the axis labels even when copying the numbers.

Point to the deciding row and column

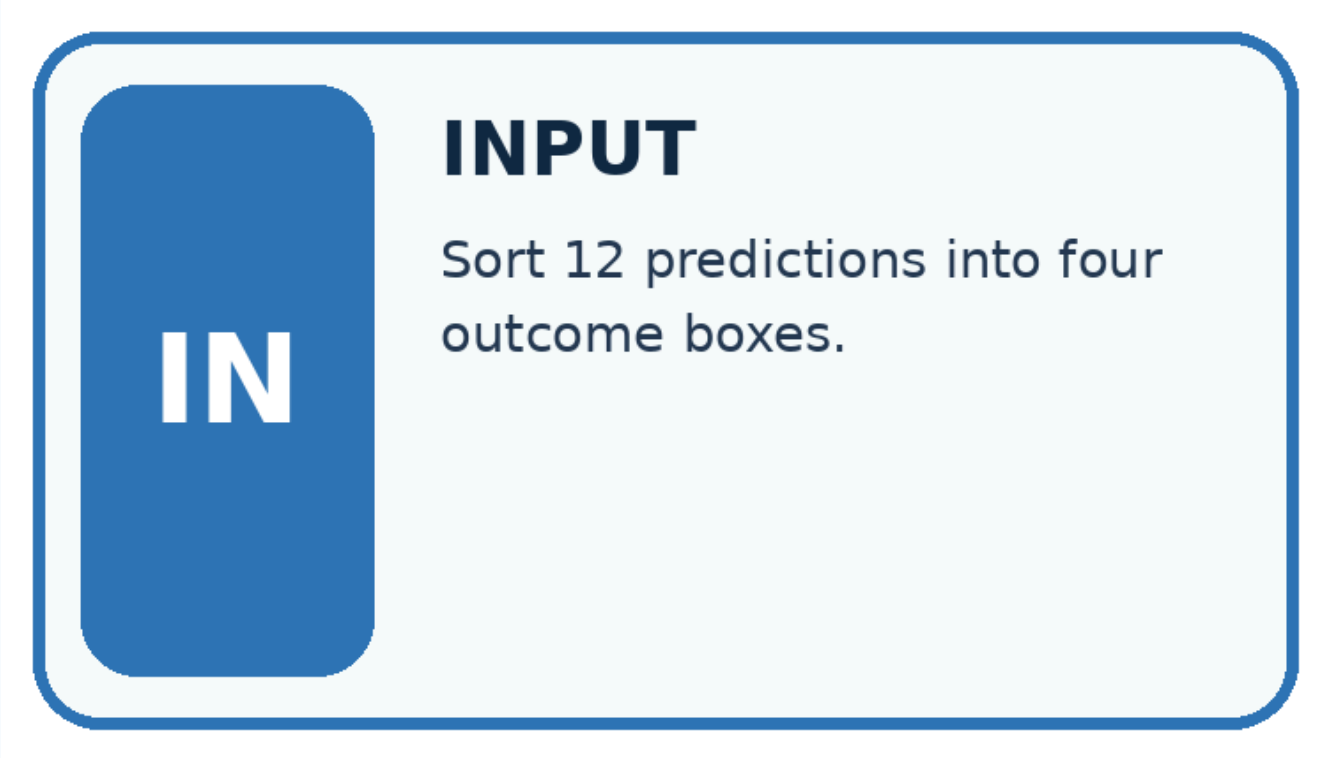
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
False alarms and misses can have different costs.
OUT
OUTPUT
Sort 12 predictions into four outcome boxes.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

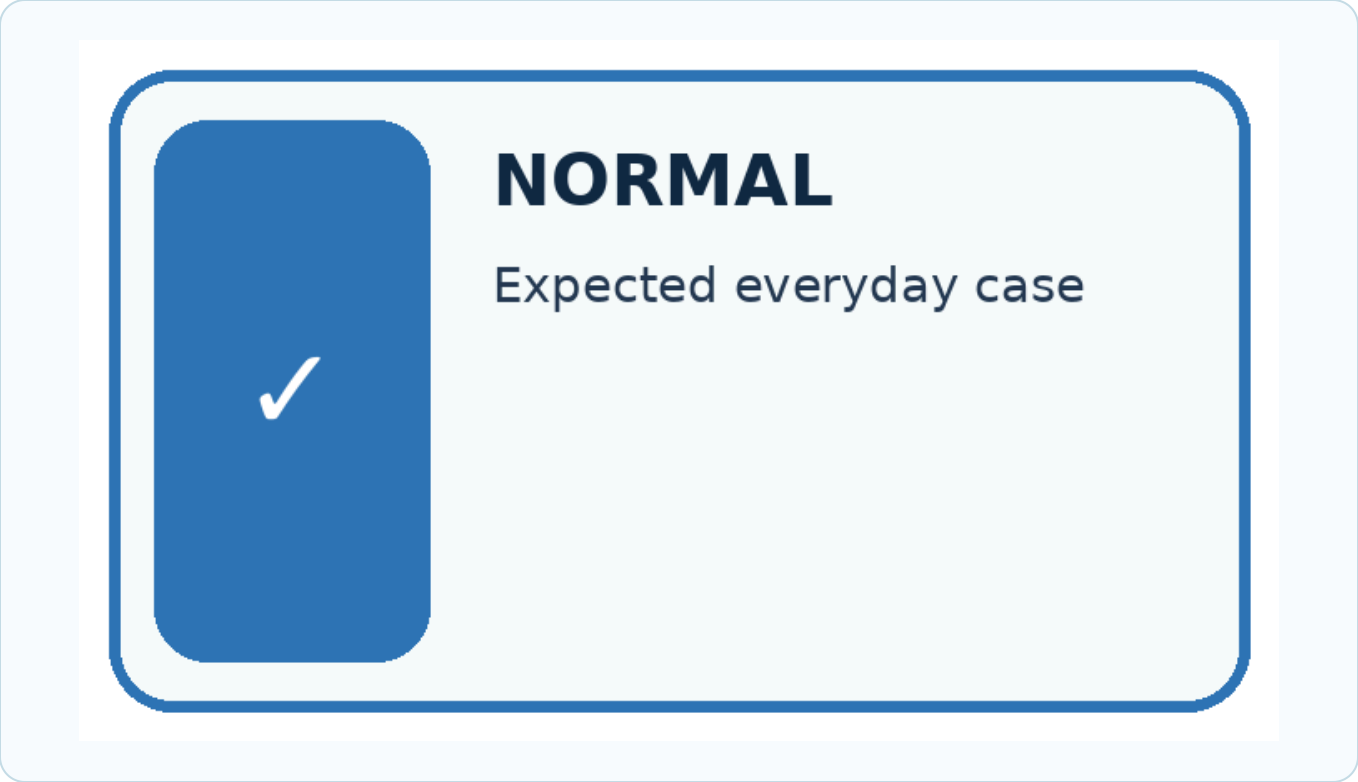
Connect the picture and the explanation

MAIN LESSON · SECTION 4

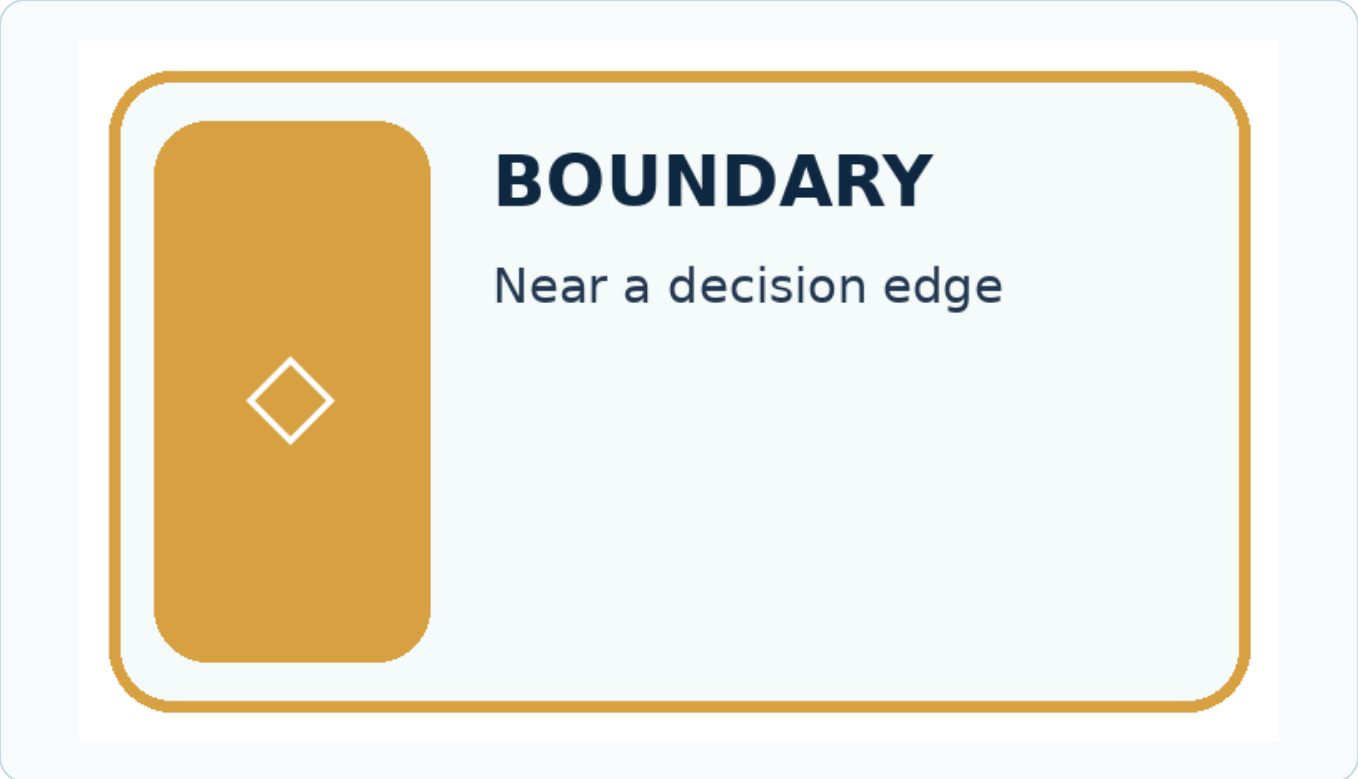
Choose a case that exposes a wrong cell definition

> Read the headers before naming an error.

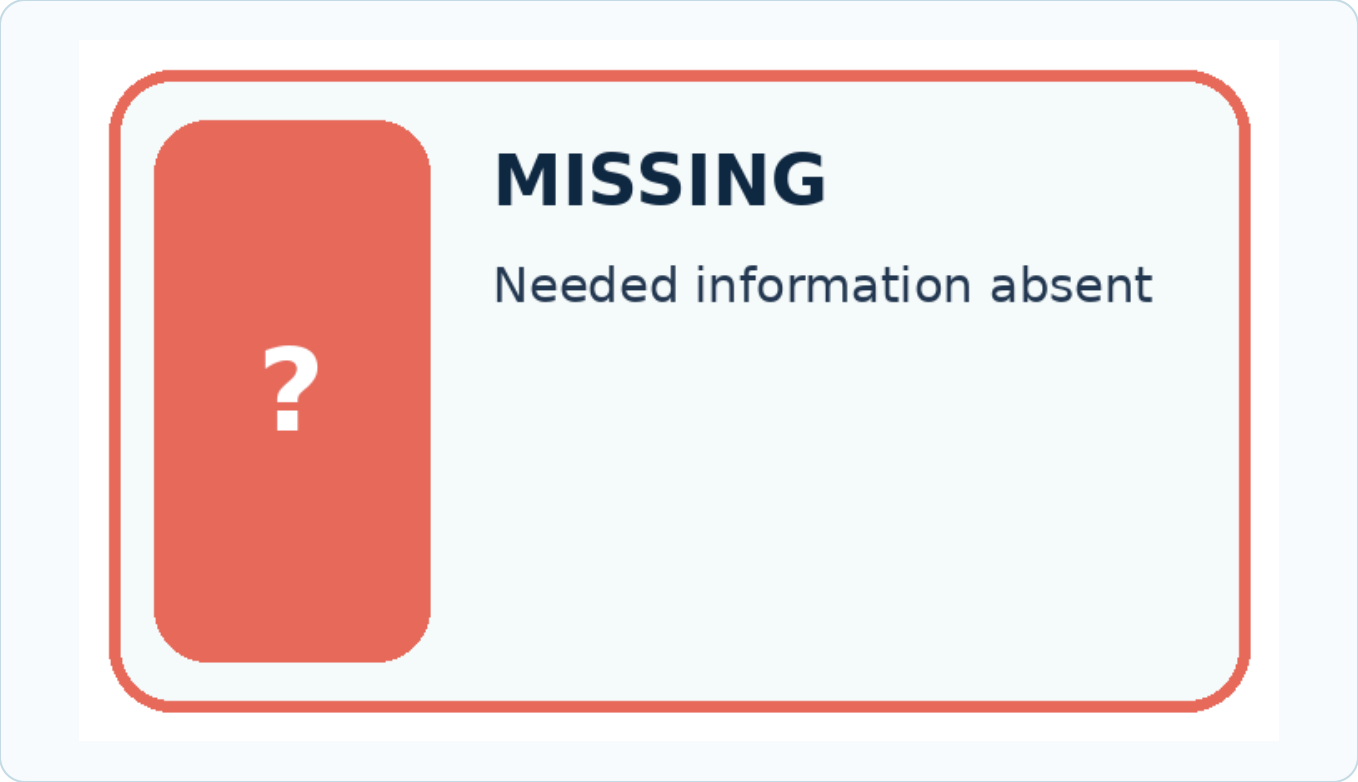
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Change one repair/sound **prediction** to repair/repair after a real model revision.

▼ Check after predicting

Prediction: Repair row becomes [0,3]; agreement becomes 5/6.

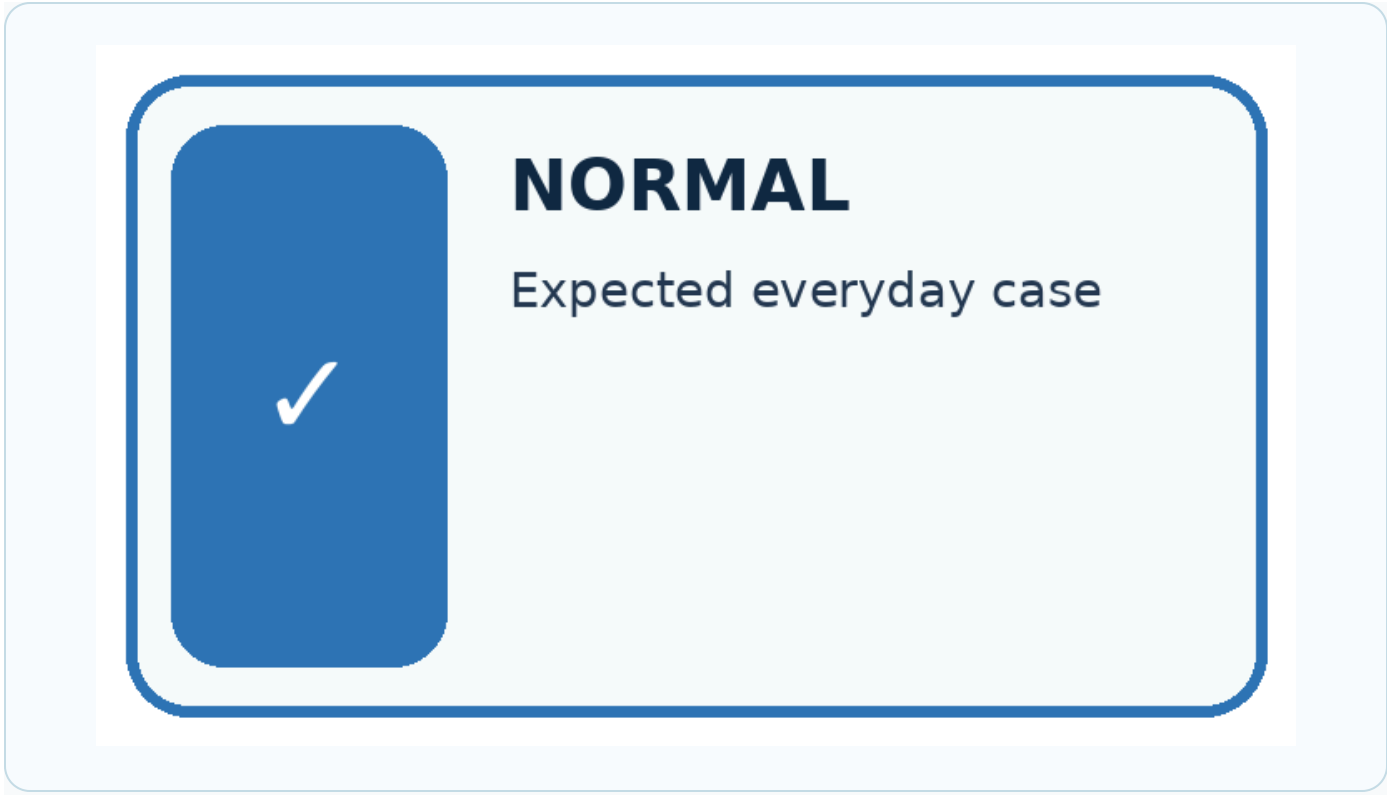
Reason: One **false negative** moves to **true positive**; the sound row stays [2,1].

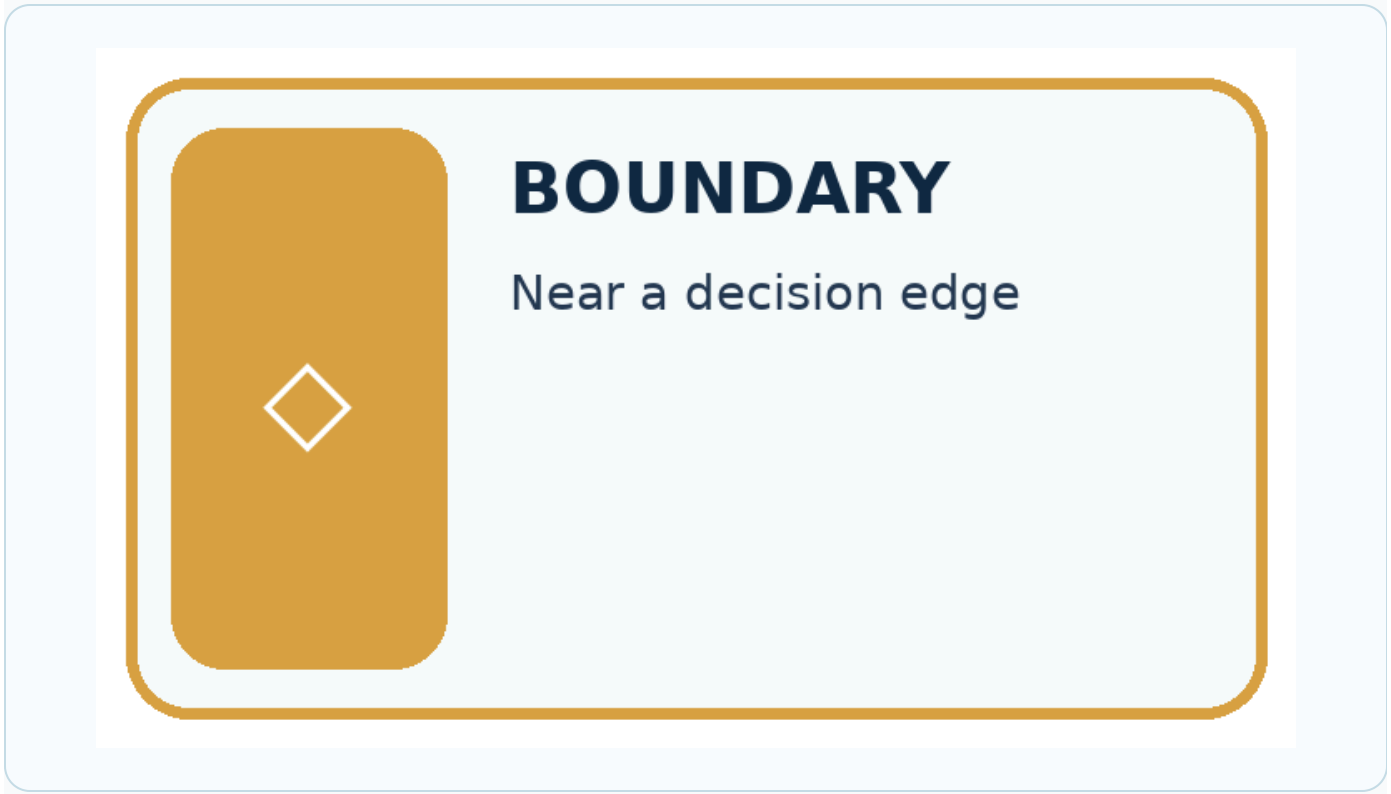
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

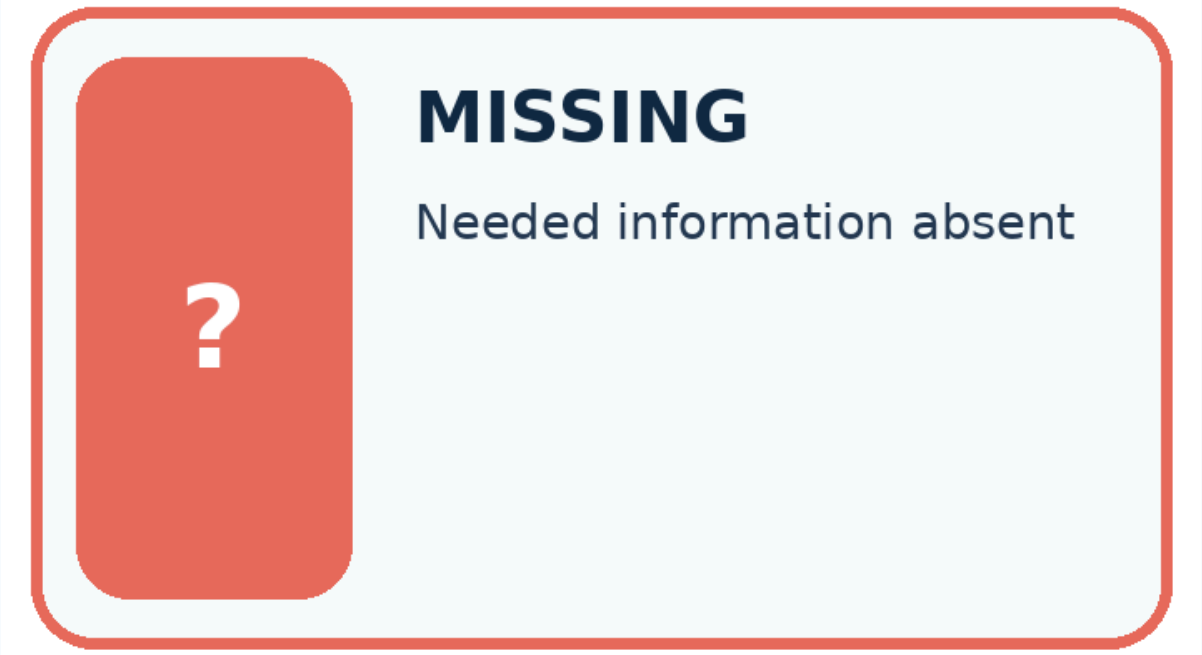
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Name the eight-record errors, then trace a separate book matrix

> The book example has six cases and uses sound/repair labels.

Match the main records to four outcome names

Blocked is the positive class

REFERENCE AND PREDICTION	CELL NAME	IDS AND COUNT
clear then clear	True negative TN	V1 V2 count 2
clear then blocked	False positive FP	V3 count 1
blocked then clear	False negative FN	V4 V5 count 2
blocked then blocked	True positive TP	V6 V7 V8 count 3

Read reference first and prediction second. Here a false positive is a false blocked alarm; a false negative is a missed blocked case.

Place V3 and V4 before discussing

> **Place V3 and V4 in the matrix. Explain which is a false positive and which is a false negative when blocked is positive.**
>
> | | | | |
> |---|---|---|---|
> | | | | |
> | | | | |
> | | | | |
> | | | | |

Keep reference uncertainty visible

True means **prediction** agrees with the reference. It does not prove that the reference annotation is infallible. Resolve disputed labels with the documented review policy rather than silently changing them to match **prediction**s.

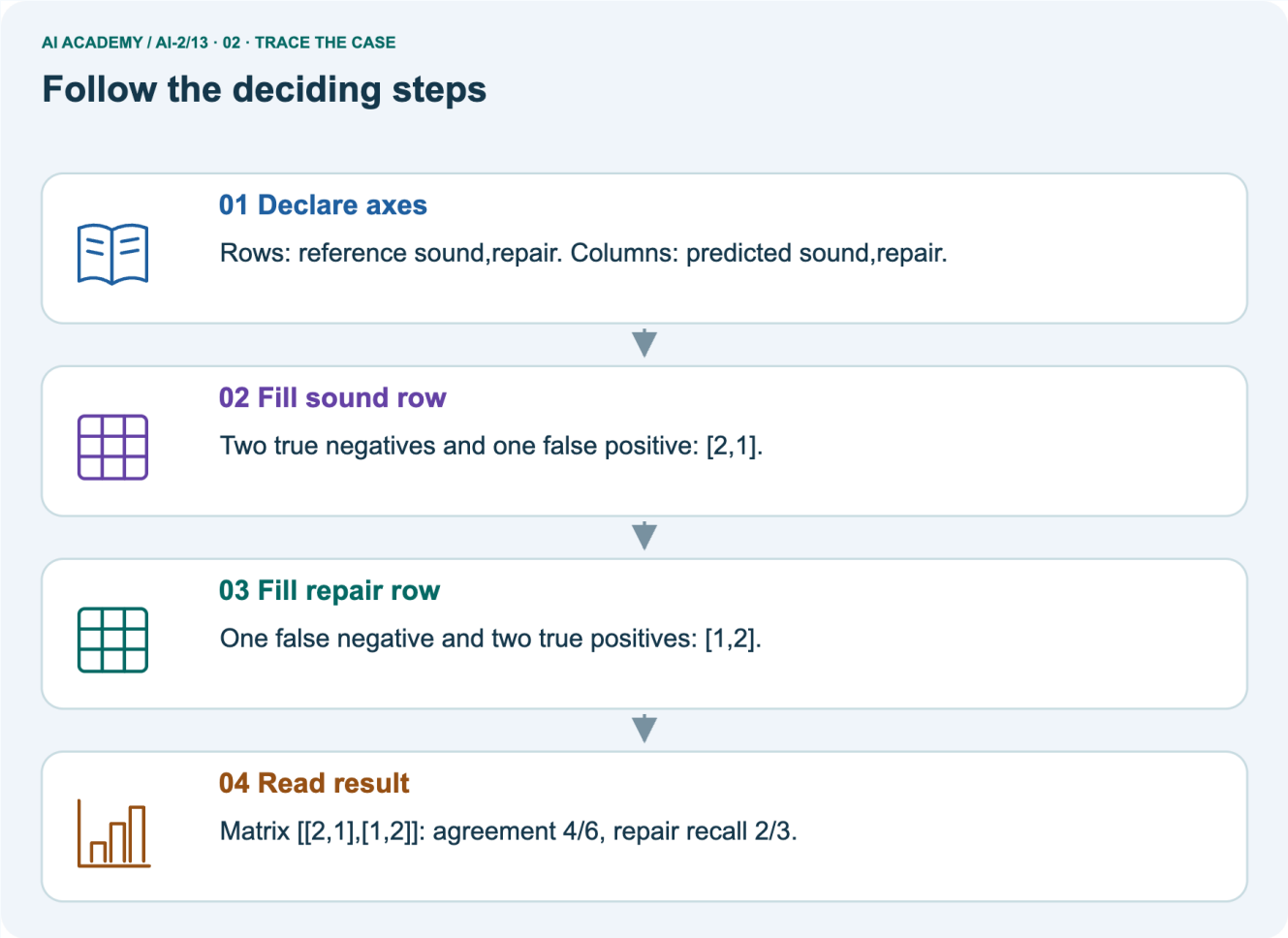
Introduce the separate six-book evaluation

A fictional book-condition checker predicts repair or sound.

Use reference rows and **prediction** columns in order sound,repair. Repair is positive.

Six reference/**prediction** pairs: sound/sound twice; sound/repair once; repair/sound once; repair/repair twice.

Trace each book-case cell



Follow the labelled picture with the matching explanation. Open the vector diagram

Each case enters one cell according to its reference and predicted label. The off-diagonal cells represent different mistakes: a false alarm on a sound book versus a missed repair.

Trace the cell definitions before naming rates; swapping axes changes the table's interpretation even if its diagonal sum remains unchanged.

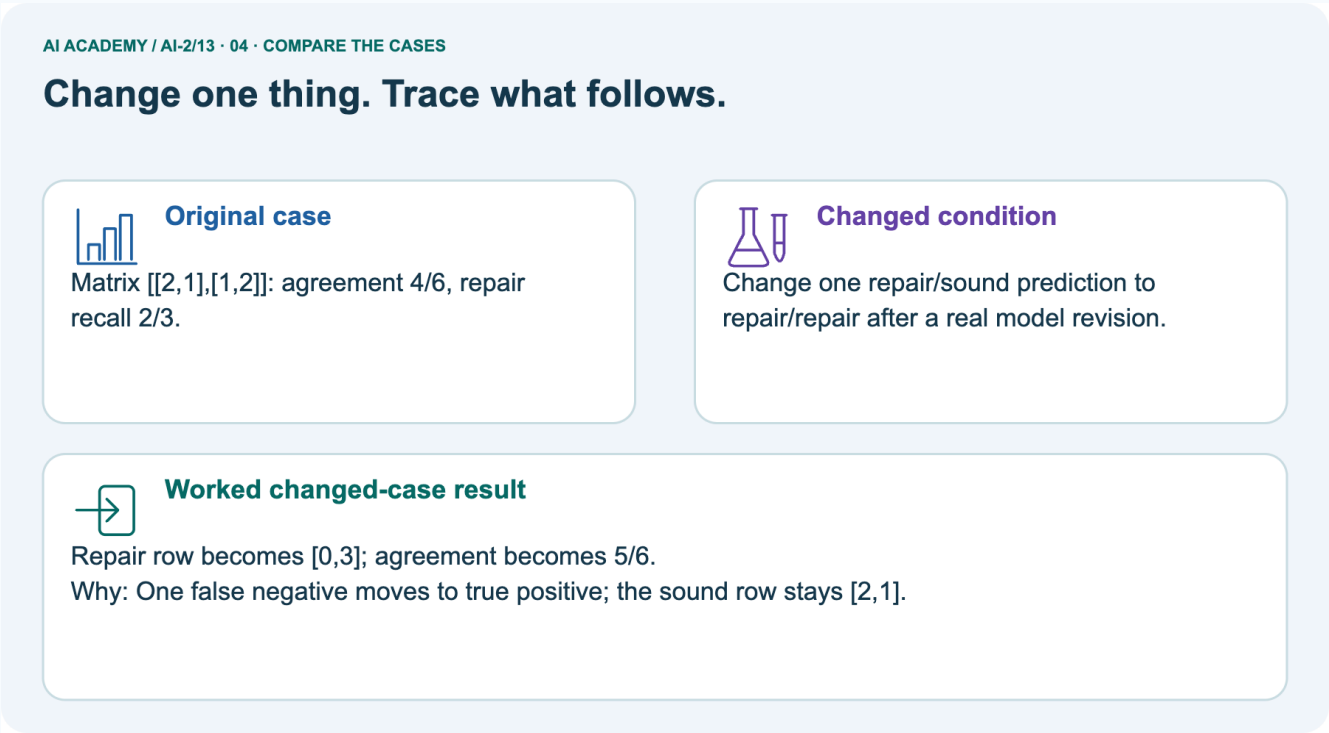
State the changed case's observation status

The changed book-case outcome is a hypothetical new-version scenario. Recounting the supplied changed prediction is an arithmetic exercise; this lesson does not run a real model revision or establish general improvement.

Predict a supplied hypothetical revision

- Change one thing: Change one repair/sound **prediction** to repair/repair after a real model revision. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the unnecessary review and label limit

- Read the labelled picture: Which error causes a sound book to be sent for unnecessary repair review?

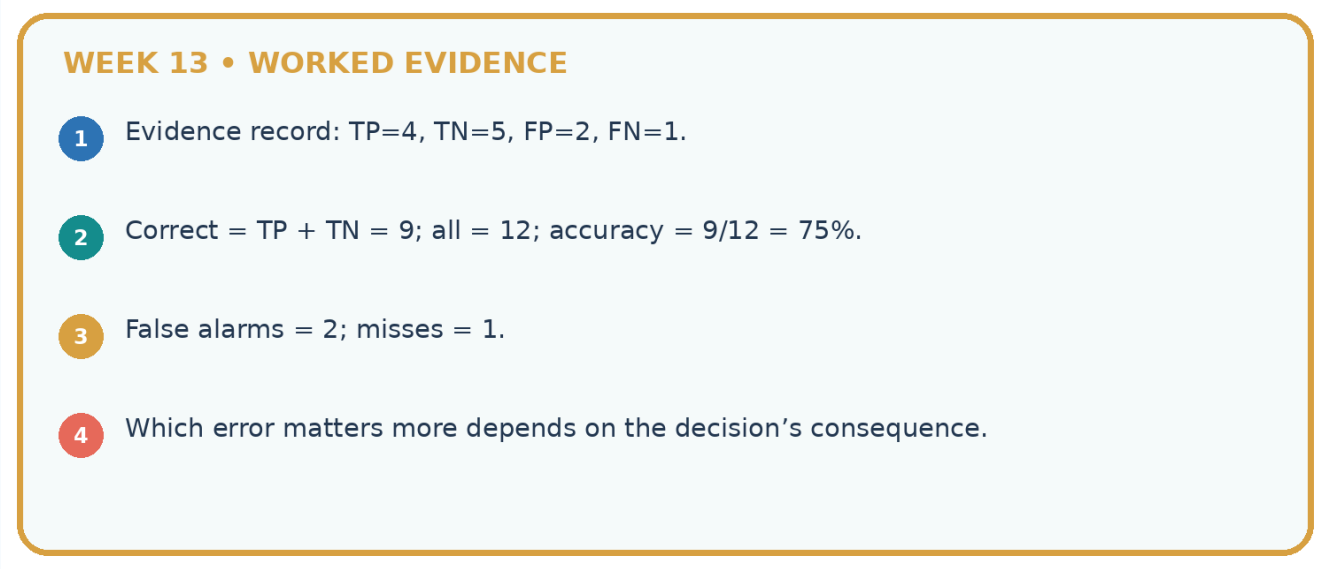
- Reference labels can themselves be disputed; a true-positive name does not guarantee the annotation is infallible.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. **Evidence** record: TP=4, TN=5, FP=2, FN=1.

2. Correct = TP + TN = 9; all = 12; accuracy = 9/12 = 75%.

3. False alarms = 2; misses = 1.

4. Which error matters more depends on the decision's consequence.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

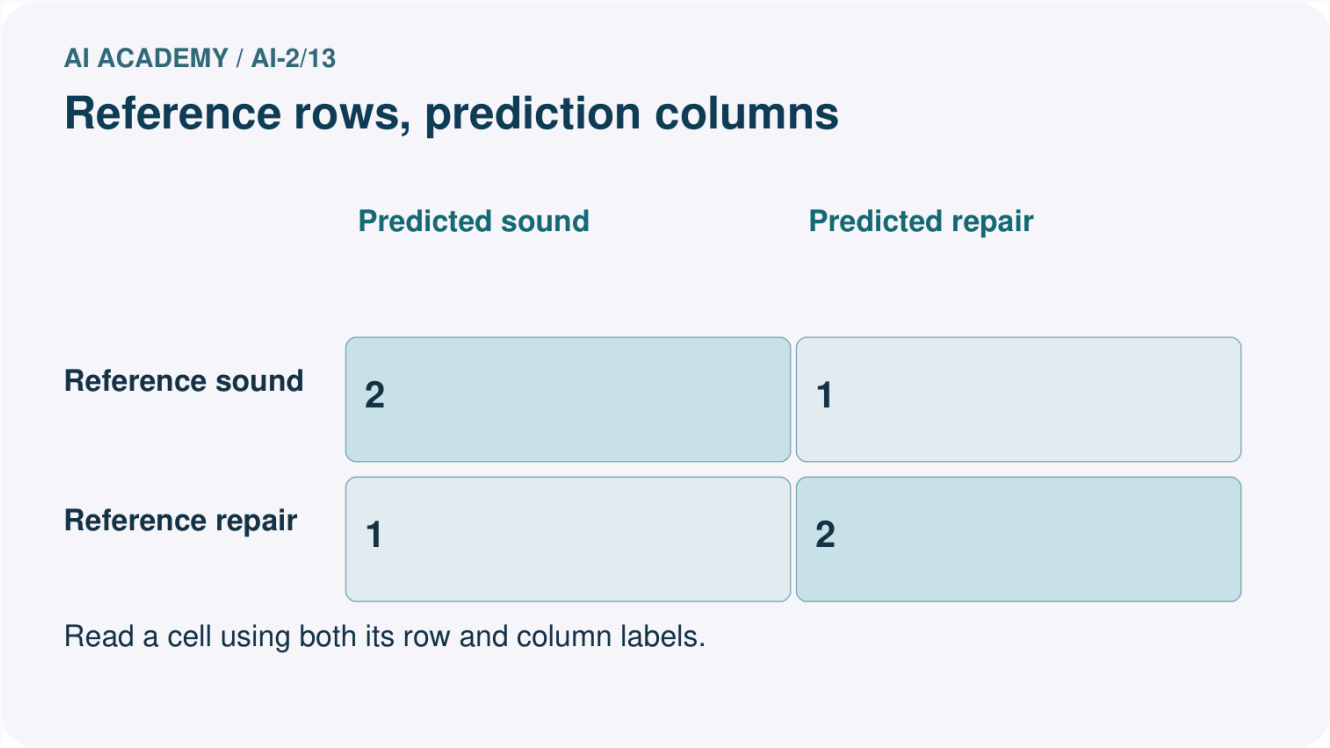
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Types of Classification Error

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/13

Types of Classification Error

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Declare axes

Rows: reference sound,repair. Columns: predicted sound,repair.

2. Fill sound row

Two true negatives and one false positive: [2,1].

3. Fill repair row

One false negative and two true positives: [1,2].

4. Read result

Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3.

CHANGE ONE THING

Change one repair/sound prediction to repair/repair after a real model revision.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Declare axes	Rows: reference sound,repair. Columns: predicted sound,repair.
2. Fill sound row	Two true negatives and one false positive: [2,1].
3. Fill repair row	One false negative and two true positives: [1,2].
4. Read result	Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

A zero cell is an observation, not a guarantee

> A matrix records error patterns without proving their cause.

Report the full axis convention and evaluation scope

Always report the evaluated split, number of records, reference and **prediction** axes, label order and **positive class**. Reversing the axis order changes positions, not the underlying outcomes.

Transposing rows and columns swaps the two kinds of error unless the labels are corrected. Read the headers before interpreting a diagram from another source.

Retain ID-level records and future limits

A zero cell means no such event in this evaluated set, not that it is impossible. A **confusion matrix** does not show the cause of errors, label quality or performance on future data. Keep the ID-level records for investigation. Next is Week14, Decision Trees. Carry forward linked records and error evidence when inspecting a tree's decisions; class imbalance was introduced in Week12.

Compare equal scores with different mistakes

> **Two classifiers have accuracy 5/8. One misses two blocked records and the other misses three. Explain how both statements can be true and what must accompany their matrices for a fair comparison.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: All mistakes should be treated as one identical error type. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check row totals, column totals and the same-record comparison

> Equal accuracy can conceal a different distribution of mistakes.

Audit the main totals and accuracy

Row totals are 3 reference-clear and 5 reference-blocked. Column totals are 4 predicted-clear and 4 predicted-blocked. All four cells sum to 8. Accuracy is the diagonal total divided by the total: 5/8, or 62.5 percent. A separate miniature three-record illustration, clear/clear, clear/blocked and blocked/blocked, has matrix [[1,1],[0,1]] and accuracy2/3. These three records are an illustration here, not the Week12 twenty-case evaluation.

Compare the alternative on the same eight references

Another classifier on these same eight reference-labelled records could have matrix [[3,0], [3,2]]. It also gets 5/8 correct, but has zero **false positive**s and three **false negative**s.

Our worked classifier has one **false positive** and two **false negative**s. Equal accuracy does not settle which error pattern is preferable; consider the task and **evidence** before choosing.

Practise outcome names in W2

Use workbook W2 for the connected example “Confusion matrices”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Repair positive means good

- Use the worked case for Confusion matrices. Proposed explanation: “Positive means good”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the separate book false alarm

Which error causes a sound book to be sent for unnecessary repair review?

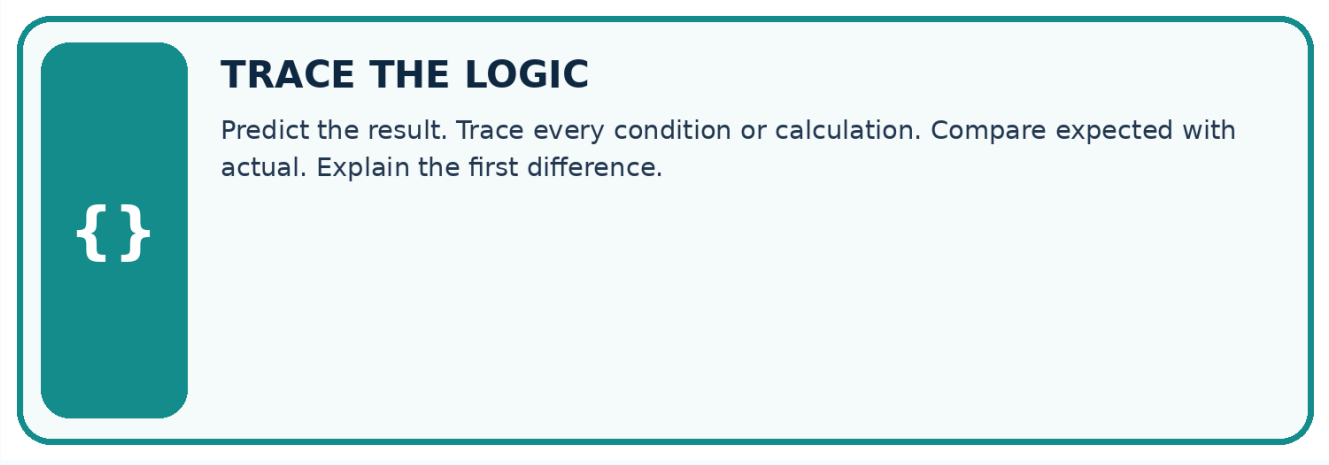
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

Practise with less help: Explain the two error types

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Build and audit the matrix from the eight supplied records

> Keep prediction, paper tally and executed output distinguishable.

Predict, place, count, check and reinterpret

1. Use the complete eight V1–V8 reference/prediction records in Section2 and the matching code in Section3. Write headers before placing each ID exactly once. Predict all four counts, row totals, column totals, diagonal total and accuracy. Run the supplied script or label the result a paper count; compare with your first tally. If there is a mismatch, find the first incorrectly placed ID rather than changing a total to match the output. Keep both attempts. Then read the same count grid with clear declared positive: keep the axes and numerical counts fixed, rename the four outcome types and explain why accuracy stays the same. No classifier is fitted, and the R1–R6 independent records remain reserved.

Check the script's input assumptions

Before counting a real evaluation, check unique IDs, allowed labels, paired references and predictions, and unresolved reviews. This supplied script assumes those checks have passed. It would count a duplicated tuple twice and raise KeyError for an unfamiliar label; neither behaviour is a complete data-review policy. Retain the source record and review a problem rather than silently dropping it or inventing a class.

Keep the evidence and one concrete consequence

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

PREDICT → TEST → RECORD → EXPLAIN

Place 12 actual/predicted pairs into a 2×2 grid, total each cell, check the total equals 12, and explain one error cost.

Place the eight supplied V1–V8 reference/prediction pairs into a labelled2×2 grid. Check all cell, row and column totals against eight records, then explain a false alarm and a missed blocked case. Keep the six R-records for independent assessment.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: count the new R1–R6 evaluation

> Build labelled cells and totals before consulting the key.

The independent paired outcomes

Each pair below is reference then **prediction**: R1 clear/clear, R2 clear/blocked, R3 clear/blocked, R4 blocked/blocked, R5 blocked/clear, R6 blocked/blocked. Use clear then blocked on both axes and reference rows.

Build the matrix, give row and column totals, name TP,TN,FP,FN with blocked positive, and calculate accuracy.

Your first attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Count a fresh evaluation independently

Week 13 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Positive means sports. A sports book predicted other and a non-sports book predicted sports occur. Name each error and its consequence for the sports shelf.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Separate recommending an unsuitable book from failing to surface a suitable one.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Positive means 'should recommend'. A book that should be recommended is rejected; a book that should not be recommended is accepted. Name both errors and explain who experiences each consequence.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Record different recommender errors and their consequences

> Keep reference policy, prediction and human decision visible.

This week’s concrete portfolio contribution

Use a declared binary category task for the fictional recommender, not an unexplained judgement that a ranking is good. Define should recommend from a documented reference policy and declare it positive. Keep paired reference/prediction outcomes by invented case ID, identify an unsuitable option shown versus a suitable option missed, and attach a concrete consequence to each. Do not infer error causes or reader enjoyment from the matrix alone.

Use a confusion table when evaluating recommender category **prediction**s and attach concrete consequences to each error.

Keep the checkpoint and evidence

THIS WEEK’S CHECKPOINT Separate recommending an unsuitable book from failing to surface a suitable one.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

Write or draw your reasoning and evidence.

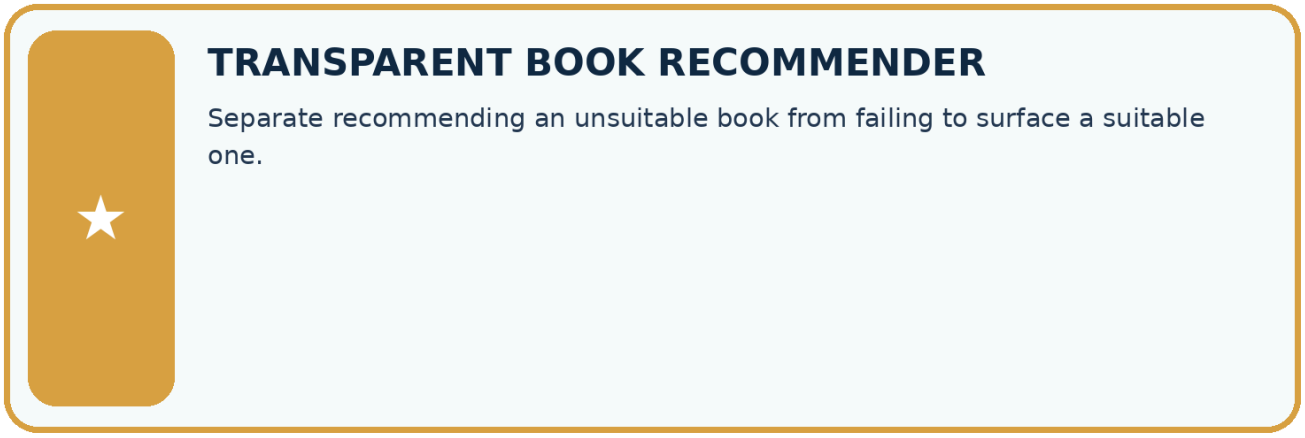
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

Finish: explain the cell and its consequence

> Name the reference, prediction and positive class before the abbreviation.

Your explanation without notes

- Close your notes. Explain this goal in your own words: Build a labelled **confusion matrix** from matched reference and **prediction** pairs

> **My deciding step, evidence and one remaining question**

Choose a changed return case

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
true positive	An actually positive case correctly predicted positive.	
true negative	An actually negative case correctly predicted negative.	
false positive	An actually negative case predicted positive, under a stated definition of positive.	
false negative	An actually positive case predicted negative, under a stated definition of positive.	

Terms for the connected explanation

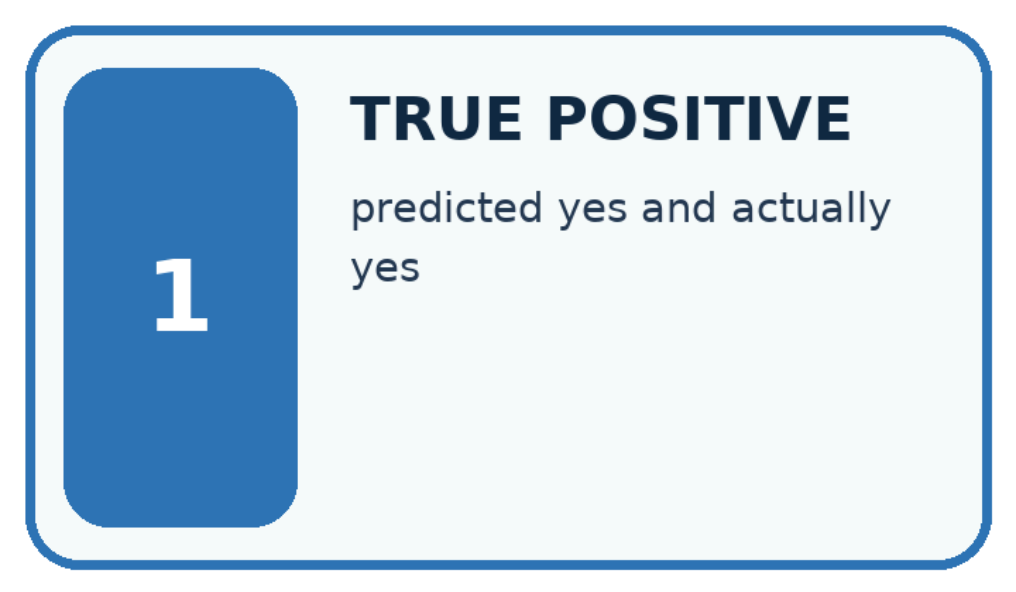
Word	Meaning
Confusion matrix	A table counting reference and prediction combinations
Positive class	The class chosen as the condition of interest

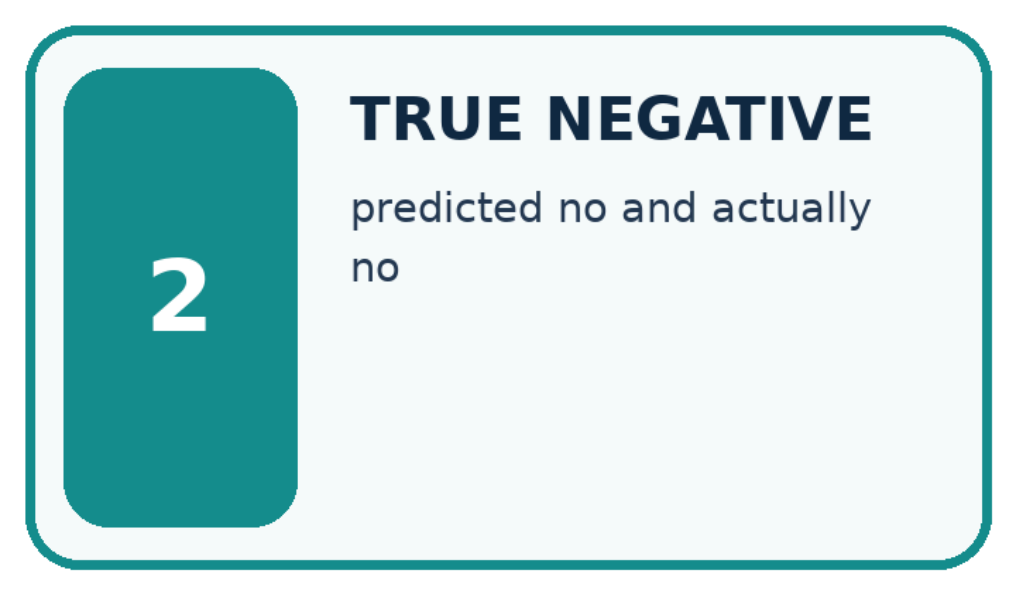
▼ **Earlier wording, alternatives and picture transcripts**

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.









Words used in the closer explanation

▼ Choose a stronger thinking challenge

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest **evidence** of independent understanding:

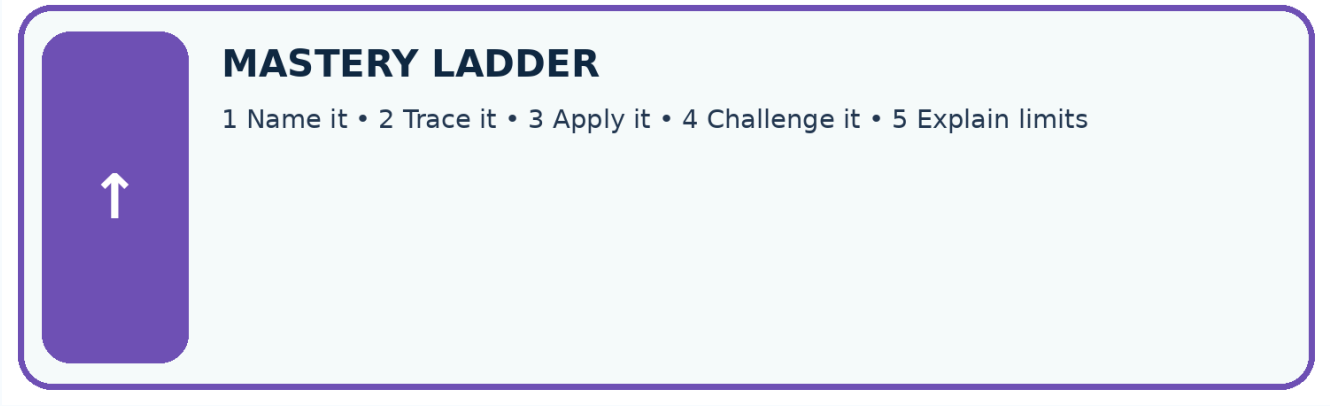
> **Write or draw your reasoning and evidence.**

▼ **Earlier wording, alternatives and picture transcripts**

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Count a fresh evaluation independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Self-check and record

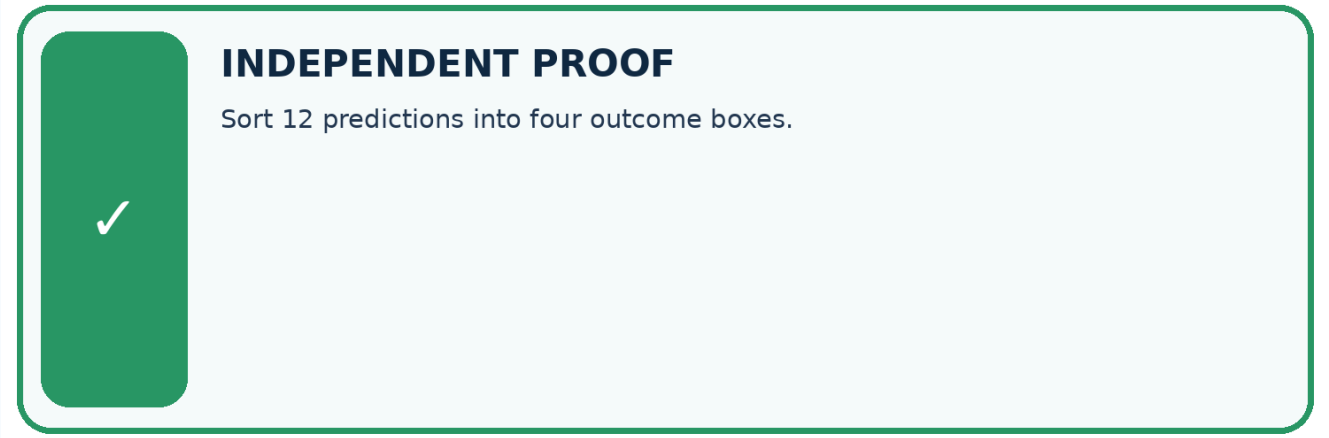
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer ‘why?’

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for explaining this week’s idea

Use after practice. Count a fresh evaluation independently is the selected independent case.



EXPLANATION CHECK Declare the axes and positive class, retain each paired ID and distinguish false alarms from missed cases.

Write or draw your reasoning and evidence.

▼ **Extra practice: Name outcomes using actual and predicted labels**

REFERENCE · SECTION 13

Extra practice: Name outcomes using actual and predicted labels

> Try the question before opening its worked explanation.

Read the supplied case and question

Positive means should recommend. Actual/predicted pairs are yes/yes, no/yes, yes/no, no/no.

Name the four outcomes.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

True positive, **false positive**, **false negative**, **true negative**. Define positive before naming errors.

Evidence or stage	Value or result
True positive	1
False negative	1
False positive	1
True negative	1

▼ Earlier wording, alternatives and picture transcripts

Week 13 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Name outcomes using actual and predicted labels

Positive means should recommend. Actual/predicted pairs are yes/yes, no/yes, yes/no, no/no.

Name the four outcomes.

True positive, **false positive**, **false negative**, **true negative**. Define positive before naming errors.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Build a labelled confusion matrix from matched reference and prediction pairs	Explain the deciding step, show a trace or test, and state one limit.
Distinguish false positives from false negatives using a declared positive class	Explain the deciding step, show a trace or test, and state one limit.
Check matrix totals and explain why equal accuracy can hide different errors	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional book-condition checker predicts repair or sound.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional book-condition checker predicts repair or sound.
- Which labelled step matches this detail? Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3.
- 1. Declare axes
- 2. Read result
- 3. Fill repair row
- 4. Fill sound row
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Read result Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3. Each case enters one cell according to its reference and predicted label.

The off-diagonal cells represent different mistakes: a false alarm on a sound book versus a missed repair.

Trace the cell definitions before naming rates; swapping axes changes the table's interpretation even if its diagonal sum remains unchanged.

Changed-case question

- Changed-case practice: Change one repair/sound **prediction** to repair/repair after a real model revision. Which conclusion is supported by the supplied facts?
- 1. Repair row becomes [0,3]; agreement becomes 5/6.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported explanation: Repair row becomes [0,3]; agreement becomes 5/6. One **false negative** moves to **true positive**; the sound row stays [2,1].

Repair and save your work

Review Accuracy as a Fraction (ai-2/12). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why is 100% repair detection not sufficient to declare the changed classifier best?.

Return to this week and explain what you changed.

Review Human Classifier Simulation (ai-2/10). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Why are J and M initially treated differently even though both are illustrated mysteries?. Return to this week and explain what you changed.

Review Rows Columns and Data Types (ai-2/7). Goal: Specify a table schema. Short practice: Which two rows originally fail, and do they fail for the same reason?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

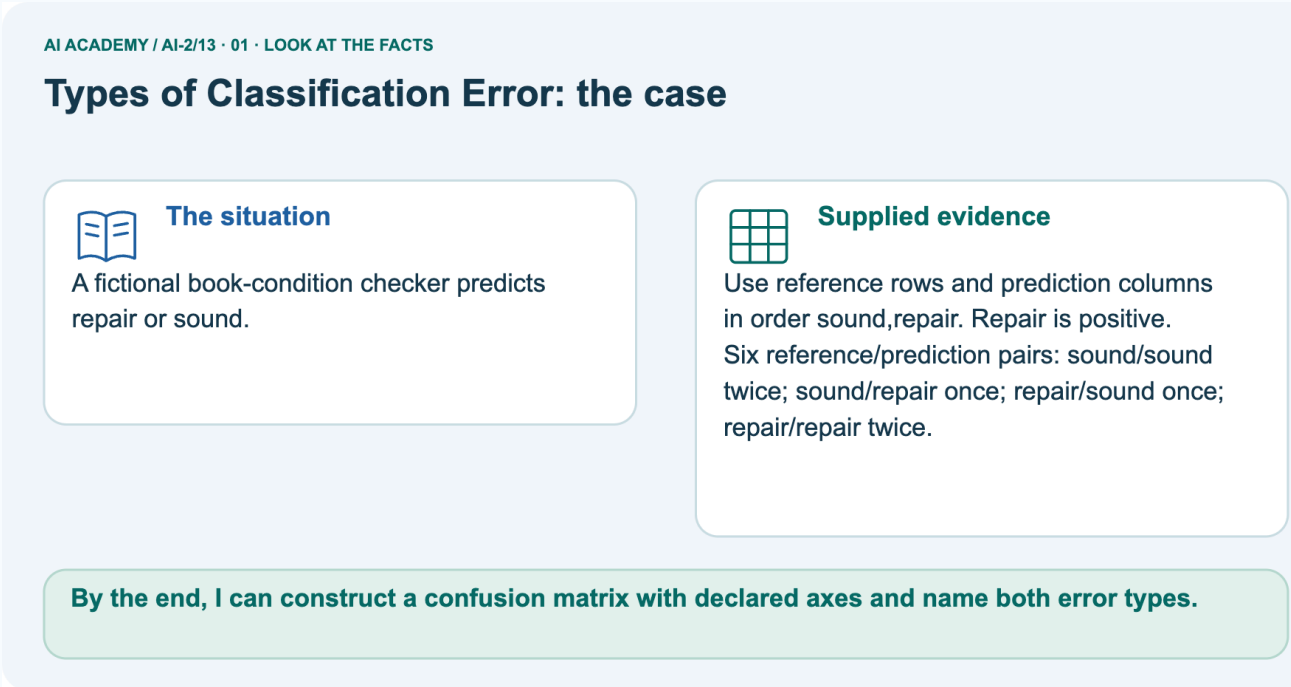
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Types of Classification Error: the case



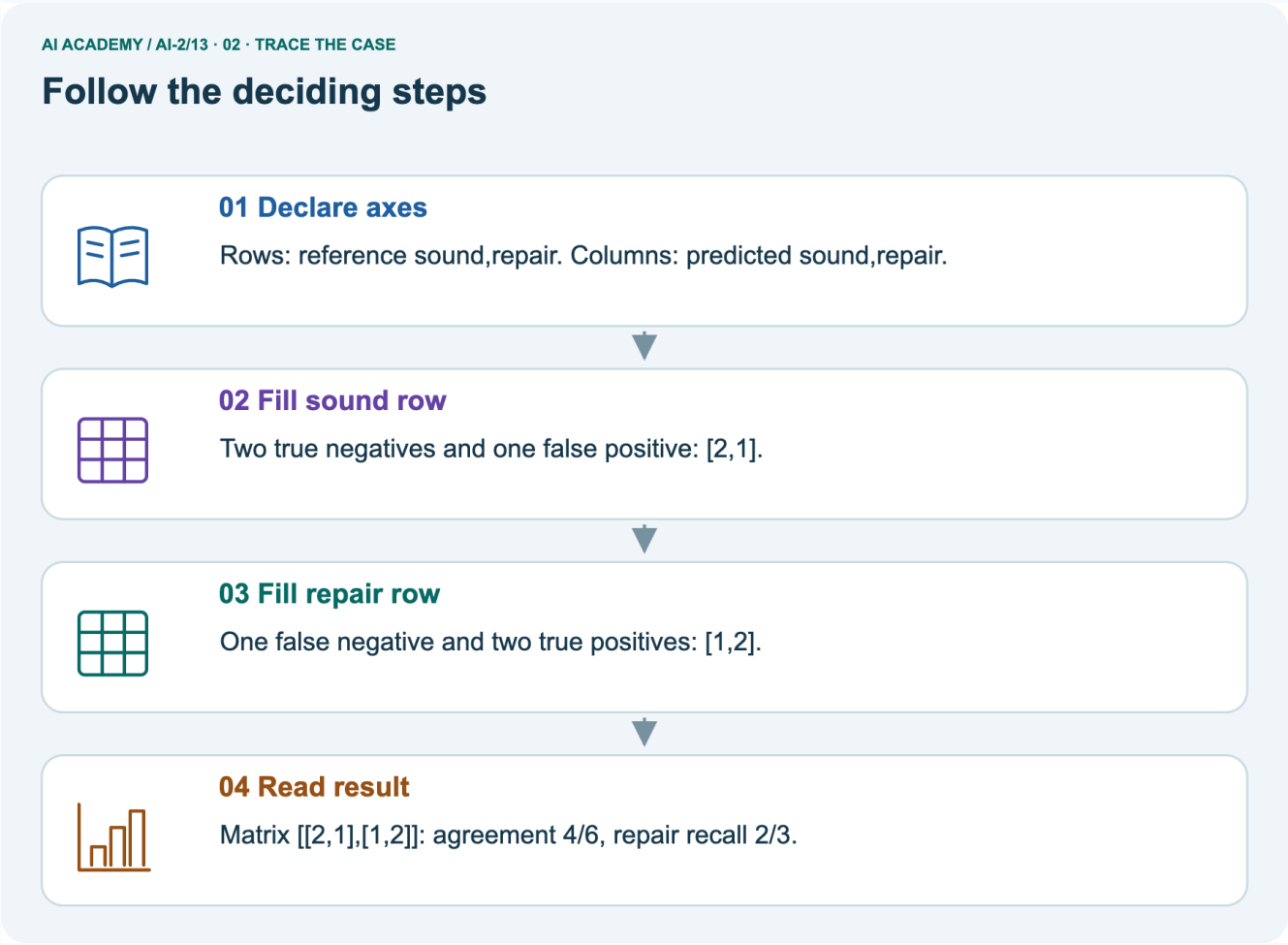
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional book-condition checker predicts repair or sound. Supplied facts: Use reference rows and **prediction** columns in order sound,repair. Repair is positive.

Six reference/**prediction** pairs: sound/sound twice; sound/repair once; repair/sound once; repair/repair twice.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



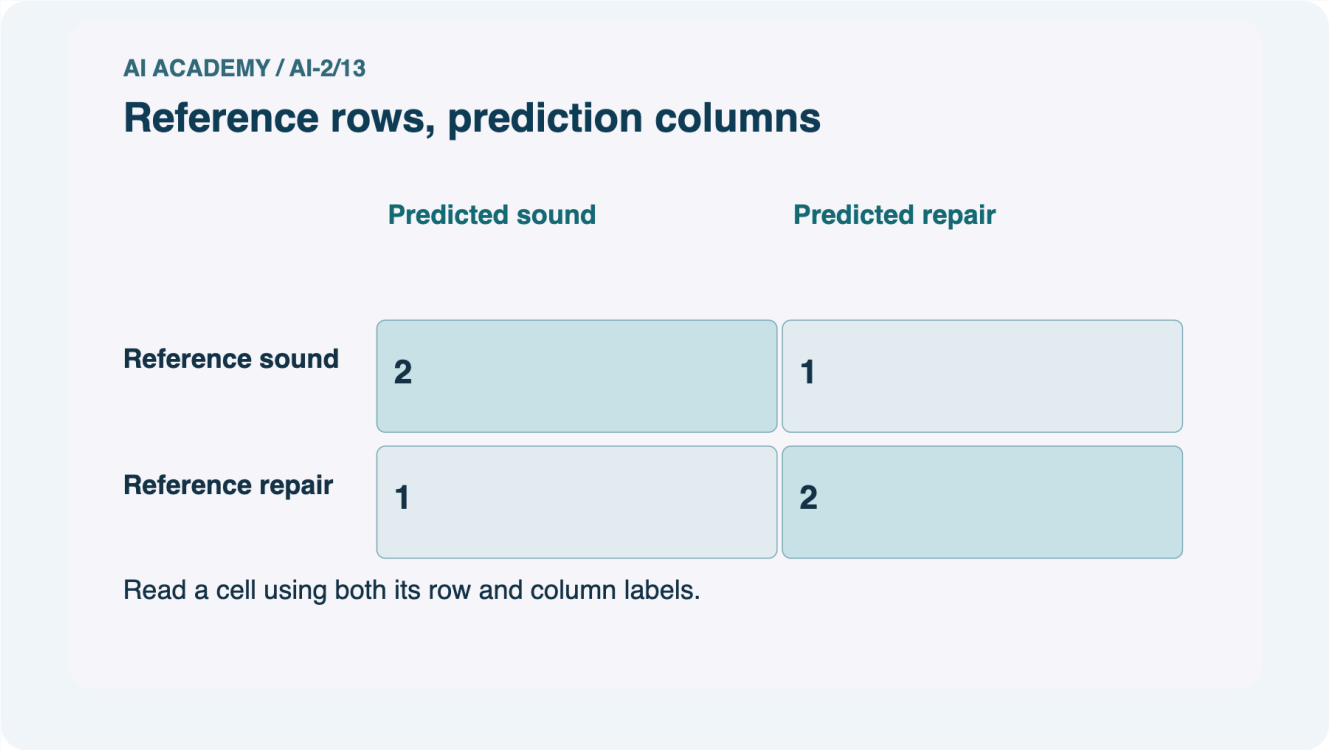
Follow the labelled picture with the matching explanation. Open the vector diagram

Declare axes: Rows: reference sound,repair.

Columns: predicted sound,repair. → Fill sound row: Two **true negative**s and one **false positive**: [2,1]. → Fill repair row: One **false negative** and two **true positive**s: [1,2]. → Read result: Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3.

Point to each numbered stage and explain which recorded **input** it uses.

03. Reference rows, prediction columns

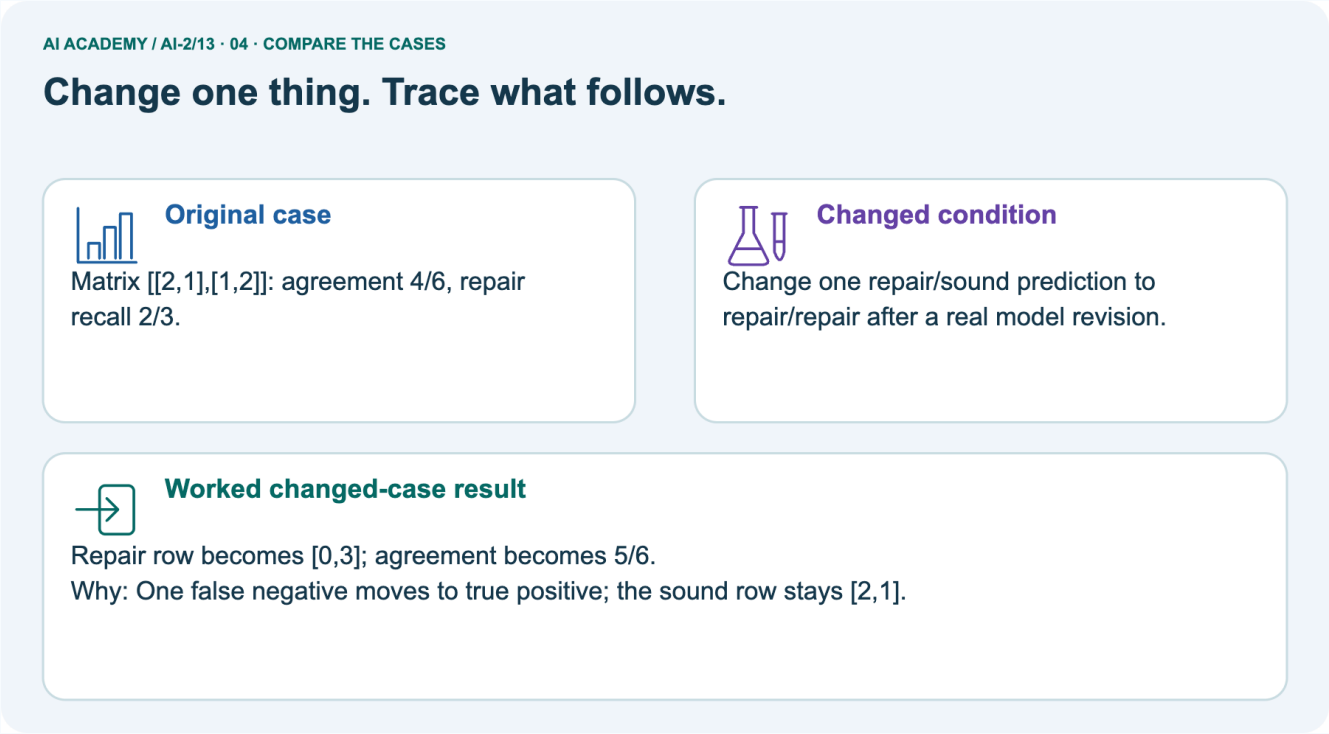


Follow the labelled picture with the matching explanation. Open the vector diagram

Reference rows, **prediction** columns. Reference sound: Predicted sound = 2, Predicted repair = 1. Reference repair: Predicted sound = 1, Predicted repair = 2.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



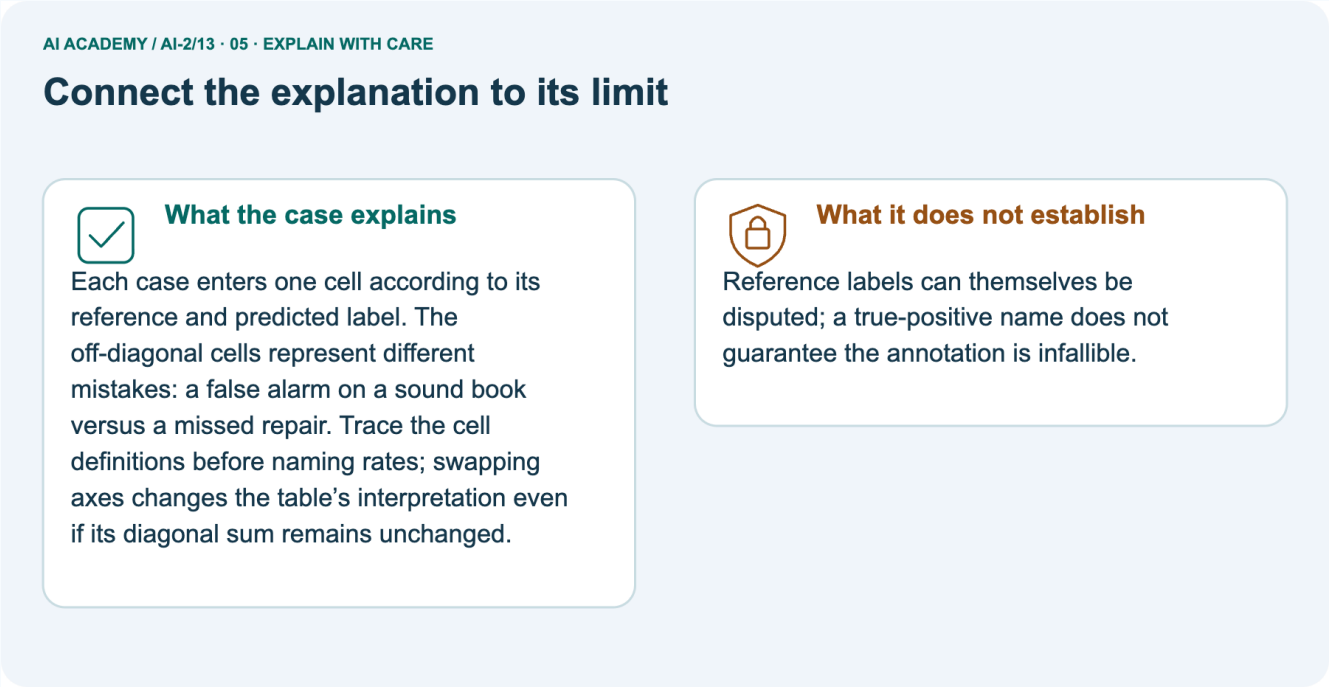
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3. Changed condition: Change one repair/sound **prediction** to repair/repair after a real model revision.

Expected result: Repair row becomes [0,3]; agreement becomes 5/6. Reason: One **false negative** moves to **true positive**; the sound row stays [2,1].

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

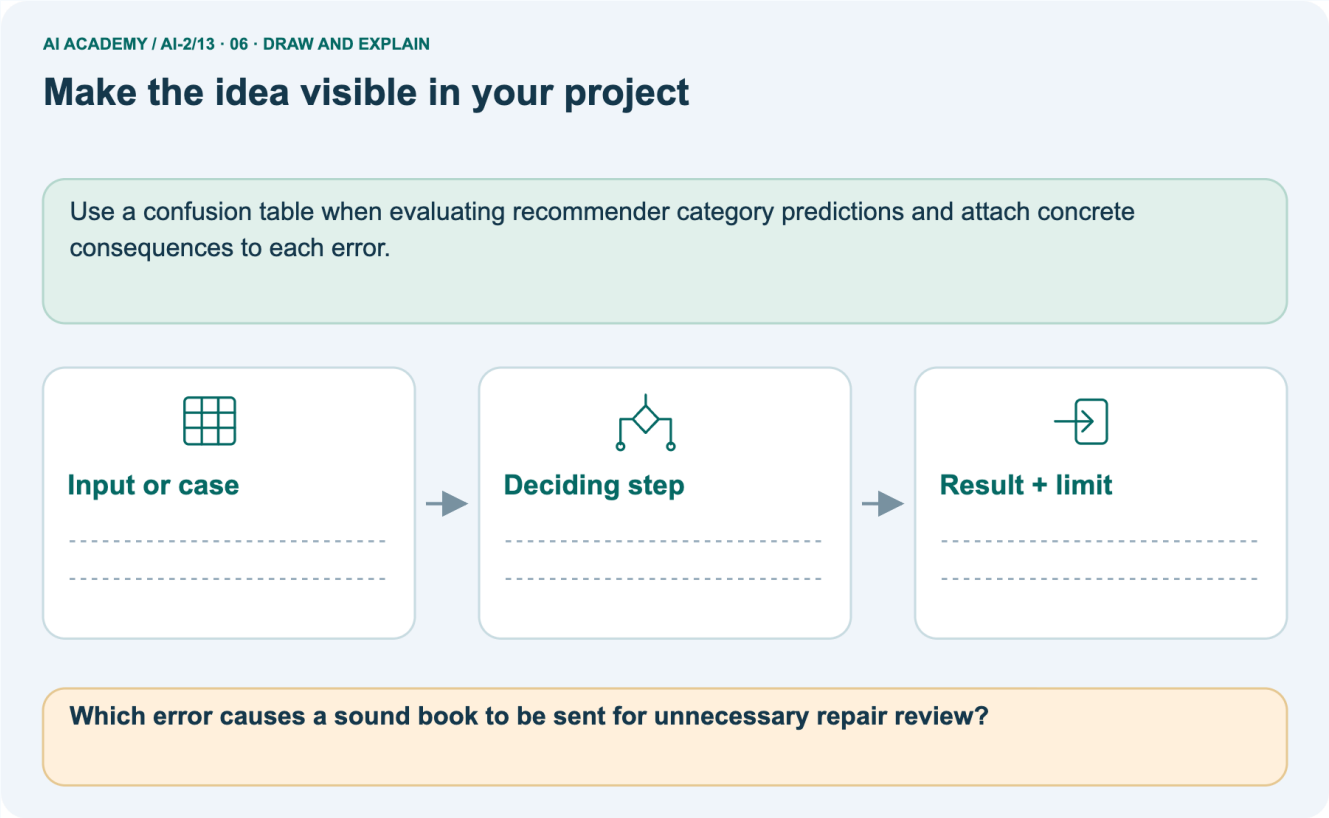
Each case enters one cell according to its reference and predicted label. The off-diagonal cells represent different mistakes: a false alarm on a sound book versus a missed repair.

Trace the cell definitions before naming rates; swapping axes changes the table's interpretation even if its diagonal sum remains unchanged.

Limit: Reference labels can themselves be disputed; a true-positive name does not guarantee the annotation is infallible.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use a confusion table when evaluating recommender category **prediction**s and attach concrete consequences to each error. Draw your **input**, deciding step, result and limit.

Supported question: Which error causes a sound book to be sent for unnecessary repair review?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Types of Classification Error: the case

02. Follow the deciding steps

03. Reference rows, prediction columns

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.